

Peifeng Yu

Curriculum Vitae

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Work Experience

- Software Engineer at Google *From Jun., 2022 to Present*
 - Tech Lead in accelerator support in Borg.
 - Design and implement generic accelerator management infrastructure in Borglet.
- Internship at Facebook *From May., 2019 to Aug., 2019*
 - Build fleet-wide GPU utilization regression detection and attribution dashboard.
 - Discover and fix data consistency issues in GPU performance data.
 - Identify optimization opportunities and give improvement suggestions via automated emails.

Education and Qualifications

2022	PhD	Computer Science and Engineering	University of Michigan
2017	MS	Computer Science and Engineering	University of Michigan
2015	B.Eng.	Software Engineering	Xi'an Jiaotong University

Publications

PhD thesis

1. Yu, P. (Aug. 2022). "Application-Aware Scheduling in Deep Learning Software Stacks". PhD thesis. University of Michigan.

Research papers

1. Yu, P. and M. Chowdhury (2020). Salus: Fine-Grained GPU Sharing Primitives for Deep Learning Applications. In: *Proceedings of Machine Learning and Systems* 2. MLSys.
2. Wesolowski, L. et al. (2021). Datacenter-Scale Analysis and Optimization of GPU Machine Learning Workloads. *IEEE Micro* **41**(5), 101–112.
3. Yu, P., J. Liu, and M. Chowdhury (2021). Fluid: Resource-aware Hyperparameter Tuning Engine. In: *Proceedings of Machine Learning and Systems* 3. MLSys.
4. Yu, P., Y. Qiu, et al. (Aug. 2022). Orloj: Predictably Serving Unpredictable DNNs. arXiv: 2209.00159. <https://arxiv.org/abs/2209.00159>.

Workshop papers

1. Nguyen, L., P. Yu, and M. Chowdhury (2017). No! Not Another Deep Learning Framework. In: *Proceedings of the 16th Workshop on Hot Topics in Operating Systems*. HotOS, pp.88–93.

Posters

1. Yu, P. and M. Chowdhury (2018a). Salus: Fine-Grained GPU Sharing Among CNN Applications. Poster presented at: SysML <https://www.sysml.cc/doc/83.pdf>.
2. Yu, P. and M. Chowdhury (2018b). Salus: Fine-Grained GPU Sharing Primitives for Deep Learning Applications. Poster presented at: Michigan AI.

Research Experience

- Orloj: Predictably Serving Unpredictable DNNs *From Feb., 2021 to Apr., 2022*

- A dynamic DNN serving system that uses empirical distributions to estimate request execution time and schedules them with efficient batching
 - Outperforms state-of-the-art serving solutions for high variance dynamic DNN workloads by 51%-80% in finish rate under tight SLO constraints.
- Fluid: Resource-aware Hyperparameter Tuning Engine *From May, 2020 to Jan., 2021*
- Generalized Hyperparameter tuning execution engine using efficient heuristics with theoretical guarantees to solve the scheduling problem.
 - Improve resource utilization by both inter- and intra-GPU training mechanisms
 - Code open sourced at <https://github.com/SymbioticLab/fluid>
- Salus: Fine-Grained GPU Sharing for Deep Learning Applications *From Sep., 2016 to Apr., 2020*
- Fine-grained GPU sharing by providing missing primitives: fast switching and memory sharing.
 - Improves GPU utilization for hyper-parameter tuning by 2.38×, and for DL inference applications by 42× over not sharing the GPU.
 - Code open sourced at <https://github.com/SymbioticLab/Salus>
- Deep Tree: SQL Injection Detection by the Power of Deep Learning *From Sep., 2016 to Mar., 2017*
- Tree-based CNN for SQL statements classification with 94.7% accuracy for injection detection.
 - Compiled new SQL statements dataset of 4161 samples.
 - Code open sourced at <https://github.com/Aetf/tensorflow-tbcnn>
- System Design of Streaming Video Analysis Application in Storm *From Apr., 2016 to July, 2016*
- Storm topology with several video classification, captioning and object tracking workloads.
 - Latency and throughput analysis on 3 GPU servers, to understand the relationship between parallelism hint and performance.
 - Summer research in Clarity Lab.
- Evaluation of Graphical Keyboard User Interface *From Sep., 2015 to Dec., 2015*
- Evaluated of two GKUI applications completing different tasks. Operating systems were also included as a variable in the experiment.
 - The result indicates significant improvements using GKUI in both tasks.
 - Course research project for Introduction to HCI Research at University of Michigan.
- Neural Network Classifier with Generalized Correntropy Loss *From Jan., 2015 to May, 2015*
- Bachelor's thesis.
 - Implemented a neural network classifier using generalized correntropy loss function.
 - Analyzed the classifier's behavior under varied order parameters in generalized correntropy loss function.
- Application data isolation using SEAndroid in NSKeyLab *From Nov., 2013 to Nov., 2014*
- Focused on the implementation of multi-domain data isolation.
 - Ported the SELinux policy compile tool chain to Android.
 - Implemented domain management and storage system service in both native user space and Android framework.
- Summer Practice in NSKeyLab (the Ministry of Education Key Lab for Intelligent Networks and Network Security on Network Traffic Capture and Analysis) *From July, 2012 to Sep, 2013*

- Mainly engaged in the development of network traffic capture and reconstruction algorithm.
- Used techniques include WinPcap, WPF, TCP stream reassembly and HTTP reconstruction.
- Analyzed 5 high speed download traffic samples and tens of HTTP traffic samples.

➤ Graphical Data Quality Management System based on IP-MAP *From Nov., 2012 to May, 2013*

- Professional graphical software offering a specialized platform for data quality management based on IP-MAP.
- Funded by the national innovation fund project of College Students of Xi'an Jiaotong University.
- Third prize by Xian Jiaotong University in the "Tengfei Cup" undergraduate curricular academic science and technology competition.
- Participated as team leader and programmer.

Project Experience

➤ Activities on GitHub *From 2012 to present*

- Contributed to several projects: KeepassXC, kmscon, Mono, CuteMarkEd, hid-apple-patched.
- Side projects including a bencode library, network traffic capture and analyze, torrent list migration between uTorrent and qBittorrent.

➤ LLDB Support for KDevelop (Google Summer of Code) *From May, 2016 to Aug., 2016*

- Extended KDevelop C/C++ debugger architecture to allow multiple debugger backends.
- Added LLDB backend for KDevelop, enabling C/C++ debugging through LLDB.
- Successful Google Summer of Code project. Link: <https://summerofcode.withgoogle.com/archive/2016/projects/6014826014834688/>
- Keep contributing to the KDevelop after the GSoC period.

➤ Contribution to open source Mozilla project *From June, 2014 to 2015*

- Fixed several bugs (features) in JavaScript JIT Engine.
- Implemented recover instruction for multiple MIR instructions.
- Got Mozillian membership.

➤ Software Low Power Mode Based on Protean Code *From Sep., 2015 to Dec., 2015*

- A runtime power saving optimization platform based on Protean Code triggered by OS power events.
- Implemented devectorization pass which disables SLP instructions when batteries are running out.
- Course project for Advanced Compiler in University of Michigan, worked in a group of 4.

➤ Pixel Cube Game for LeapMotion *From Nov., 2013 to June, 2014*

- 3D pixel painting using hand gestures, powered by LeapMotion
- Course project for Software Project Management in Xi'an Jiaotong University, worked in a group of 8.

➤ Mouse Control with Kinect *From May, 2012 to July, 2012*

- Developed algorithm to smooth mouse movement by projecting hand movement onto a cylindrical surface.
- The algorithm can adapt to different body parameters.

Awards and Scholarships

- 2014 Google Excellent Scholarship (\$1500, only one in Xi'an Jiaotong University)
- 2014 Meritorious Winner of Interdisciplinary Contest in Modeling
- 2013 Silver Medal of the ACM-ICPC Asia China
Shaanxi Provincial Programming Contest
- 2013 Fuji Xerox (China) Scholarship (RMB5000, 4 of top 20% student in the major)
- 2013 Merit Student in Xi'an Jiaotong University
- 2012 First prize for MCM/ICM of Xi'an Jiaotong University
- 2012 Third prize for ACM Programming Contest of Xi'an Jiaotong University
- 2012 Third prize for "Tengfei Cup" Undergraduate
Curricular Academic Science and Technology Competition
- 2011 Excellent Student Cadre in Xi'an Jiaotong University
- 2011 "Lu Shidi" Scholarship (RMB6000, 2 of top 10% students in the major)